Practice workbook · Find a mean, median, mode or mean rate of change

A set of repeat readings or survey values needs an average, or a change over a time interval needs a mean rate per unit time.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-MEAN-MEDIAN-MODE-RATE

1. Question 5 · part a.i · 2 marks

June 2019 Paper 1

Worked-example retry · Attempt part a.i only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

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mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

.....

2 .....

.....

(iii) Give two ways the teacher could improve his investigation.

(2)

1 .....

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2 .....

.....

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1. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 7 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

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2. Question 7 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

Number of bubbles of CO_2	Time taken for chymosin to turn milk protein solid, in seconds				
	Test 1	Test 2	Test 3	Test 4	Mean
100	253	257	250	260	255
150	238	232	241	229	
200	216	214	219	211	215
250	208	202	212	198	205
300	210	200	199	311	203

(i) State three factors that the student should control.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

(2)

mean time taken = ...

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P 6 2 0 5 4 A 0 2 0 2 4

2. Question 7 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) In test 4 there is an anomalous result.

State how the student deals with this result.

(1)

.....

.....

(iv) State why recording the number of bubbles may produce inaccurate results.

(1)

.....

.....

(Total for Question 7 = 14 marks)

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Working and answer

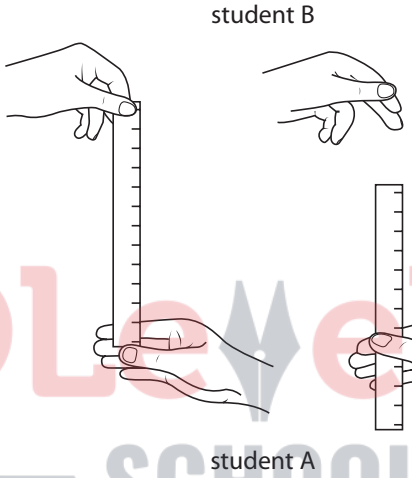
3. Question 5 · part a · 2 marks

November 2021 Paper 2

Further practice · Attempt part a only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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3. Question 5 · part a · 2 marks

November 2021 Paper 2

Further practice · Attempt part a only.

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The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

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(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

3. Question 5 · part a · 2 marks · Working and answer

Working and answer

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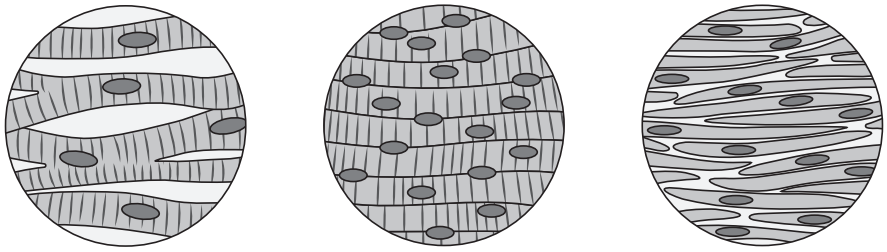
4. Question 6 · part b.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

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6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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4. Question 6 · part b.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

74%	75%	70%	82%	74%	74%	78%	82%	70%	82%
78%	74%	82%	78%	82%	75%	84%	72%	82%	72%

(i) Calculate the mean percentage muscle mass.

(2)

mean percentage muscle mass = ...%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

(3)

(iii) Determine the mode of these percentage muscle masses.

(1)

mode = ...

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P 7 5 9 5 8 A 0 1 6 2 4

4. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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5. Question 7 · part c.ii · 2 marks

June 2022 Paper 1

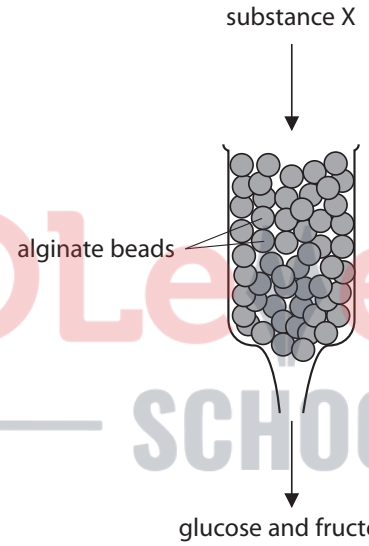
Further practice · Attempt part c.ii only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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5. Question 7 · part c.ii · 2 marks

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Further practice · Attempt part c.ii only.

(b) Which process is involved in the conversion of glucose into glycogen in the human body?

(1)

- ☒ **A** glucagon is released from the pancreas and travels to the liver
- ☒ **B** insulin is released from the pancreas and travels to the liver
- ☒ **C** glucagon is released from the liver and travels to the pancreas
- ☒ **D** insulin is released from the liver and travels to the pancreas

(c) (i) The total population of the United Kingdom is 66.65 million.

Diabetes affects 6.0% of this population.

Calculate the number of people with diabetes in the United Kingdom.

(2)

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number of people = ... million

(ii) Surveys were carried out in different areas of the United Kingdom.

These surveys collected data on the number of people in each area with diabetes.

The information collected is shown in the box.

38	62	51	96	49	78	82	91	65

Determine the median number of people with diabetes.

(2)

median = ...

(Total for Question 7 = 10 marks)

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5. Question 7 · part c.ii · 2 marks · Working and answer

Working and answer

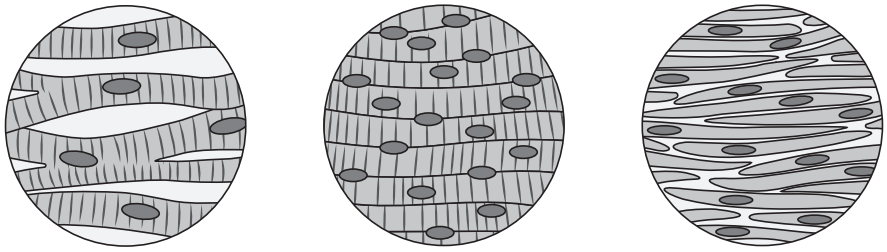
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6. Question 6 · part b.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part b.iii only.

6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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6. Question 6 · part b.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part b.iii only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

74%	75%	70%	82%	74%	74%	78%	82%	70%	82%
78%	74%	82%	78%	82%	75%	84%	72%	82%	72%

(i) Calculate the mean percentage muscle mass.

(2)

mean percentage muscle mass = ...%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

(3)

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(iii) Determine the mode of these percentage muscle masses.

(1)

mode = ...

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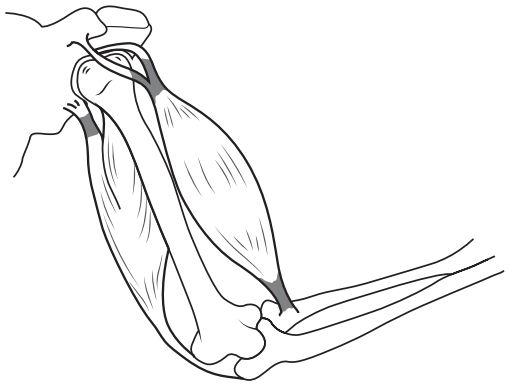
16

6. Question 6 · part b.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part b.iii only.

(c) The diagram shows two muscles in the arm.



Explain how these muscles help to raise the forearm.

(4)

(Total for Question 6 = 17 marks)

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6. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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7. Question 4 · part a.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part a.ii only.

4 A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

1.00 au = −7 %	1.50 au = −10 %	0.00 au (distilled water) = +8 %
	0.25 au = +4 %	0.50 au = −1 %

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

(4)

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P 7 8 7 6 9 A 0 8 2 0

7. Question 4 · part a.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part a.ii only.

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(ii) Determine the median value for sugar concentration.

(1)

median value = ...

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

(3)

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mass of cylinder = ... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

(4)

concentration of sugar solution = ... au

7. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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8. Question 3 · part a.i · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a.i only.

3 A woman who smokes when she is pregnant risks harming her developing fetus.

The table lists the birth masses of 310 children born to non-smoking mothers or to mothers who smoke.

	Birth masses of children in kg					
	Less than 2.4	**2.4 – 3.1**	**3.2 – 3.5**	**3.6 – 3.9**	**More than 3.9**	**Total number of births in sample**
Number of children born to non-smoking mothers	4	42	70	60	34	210
Number of children born to mothers who smoke	3	24	44	23	6	100

(a) (i) What is the most common birth mass in kg? **(1)**

- ☒ **A** 2.4 – 3.1
- ☒ **B** 3.2 – 3.5
- ☒ **C** 3.6 – 3.9
- ☒ **D** More than 3.9

(ii) Calculate the percentage of children with a birth mass of 3.6 kg or more in the sample of non-smoking mothers. **(2)**

percentage = ...

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P 6 0 1 8 7 A 0 8 2 4

8. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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9. Question 7 · part b.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

7 Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 cm^3 of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 cm^3 of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

Petri dish	Diameter of bacterial growth in mm				
	day 1	day 2	day 3	day 4	day 5
A	0	1.7	3.0	4.6	7.1
B	0	4.2	8.4	15.1	36.5
C	0	0	0	0	0

(a) (i) State three control variables in the student's investigation.

(3)

1 .....

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2 .....

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3 .....

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9. Question 7 · part b.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

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(ii) State the dependent variable in the student's investigation.

(1)

.....

(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

(2)

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

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(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

(4)

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

(2)

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9. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

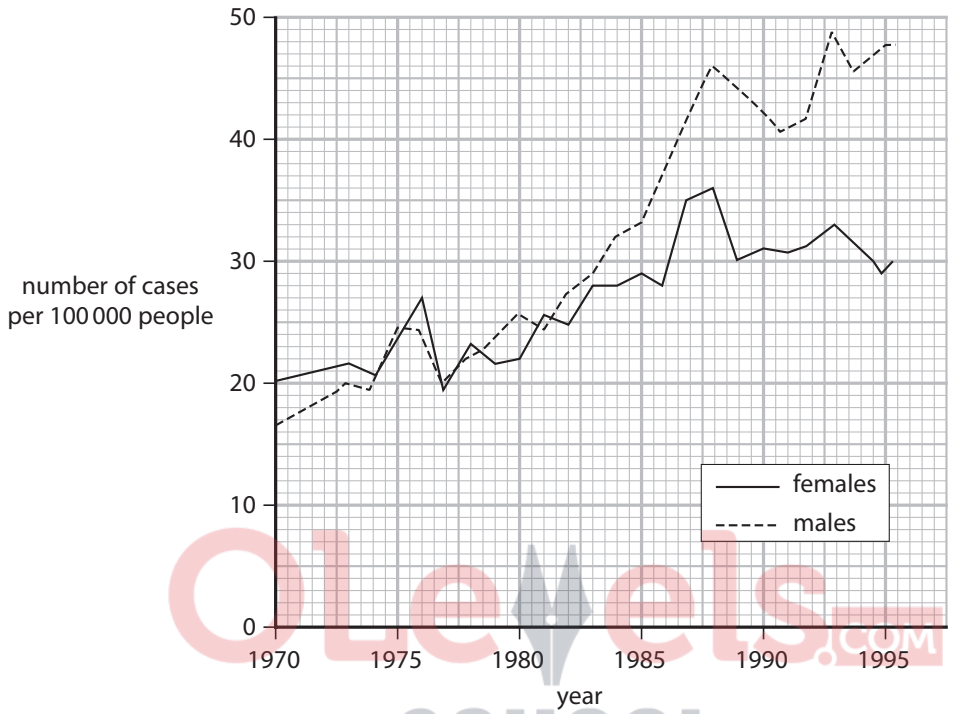
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10. Question 6 · part b.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.i only.

6 A form of skin cancer called melanoma has been increasing in both males and females.



(a) (i) Describe the trends shown by the graph.

(3)

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10. Question 6 · part b.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.i only.

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(ii) Suggest why cancer can occur in the skin.

(2)

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(b) (i) Calculate the mean increase in the number of melanoma cases per 100 000 males per year between 1970 and 1995.

(3)

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mean increase per 100 000 males per year = ...

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(ii) In 1970, the population of this area of the world was 7 million.

Calculate the number of females who had melanoma.

(3)

number of females with melanoma = ...

(Total for Question 6 = 11 marks)

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10. Question 6 · part b.i · 3 marks · Working and answer

Working and answer

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11. Question 7 · part b.i · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease?

(1)

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics.

(2)

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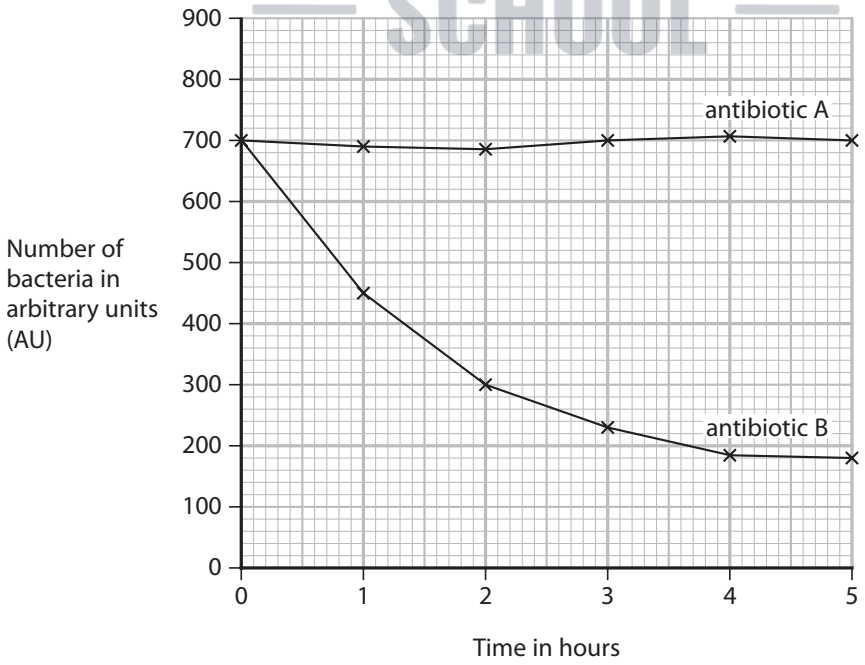
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



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11. Question 7 · part b.i · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

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(i) Calculate the mean rate of change of the bacterial population for antibiotic B during the five hours.

Give your answer to 2 significant figures.

(3)

mean rate of change = ... AU per hour

(ii) A bactericidal antibiotic kills bacteria.

A bacteriostatic antibiotic prevents bacteria from reproducing.

Explain which antibiotic is bactericidal.

Use data from the graph to help your answer.

(2)

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11. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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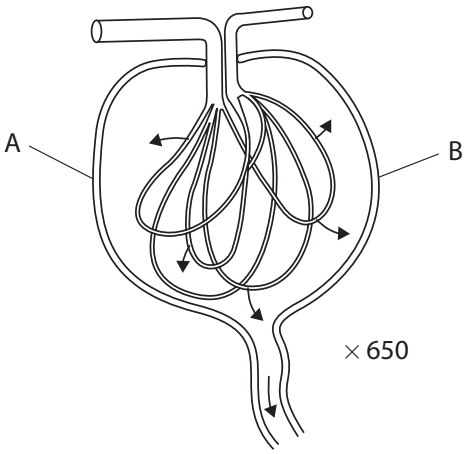
12. Question 1 · part b.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

12. Question 1 · part b.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

Time after drinking in hours	Volume of urine in cm^3	
	day 1 (20 °C)	day 2 (30 °C)
0.5	70	50
1.5	175	70
2.0	300	200
2.5	250	190
3.0	210	150
4.0	100	50

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12. Question 1 · part b.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

(i) State two factors that should be controlled in the investigation.

(2)

1 .....

.....

2 .....

.....

(ii) The mean rate of urine production on day 1 is 276 cm^3 per hour.

Calculate the mean rate of urine production on day 2 during the 4 hour time period.

(2)

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mean rate of urine production on day 2 = ... cm^3 per hour

(iii) Explain the difference between the mean rate on day 1 and the mean rate on day 2.

(3)

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(Total for Question 1 = 16 marks)

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12. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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